

Module 3

(6 hrs)

Module	Detailed Content	References	Hours/Weight
3	Social Media Text, Action & Hyperlink Analytics	T1- Ch 3,5,7	
	Social Media Text Analytics - Types of Social Media Text, Purpose of Text Analytics, Steps in Text Analytics, Social Media Text Analysis Tools Social Media Action Analytics - What Is Actions Analytics? Common Social Media Actions, Actions Analytics Tools Social Media HyperLink Analytics - Types of Hyperlinks, Types of Hyperlink Analytics, Hyperlink Analytics Tools		6 (20%)

Module 3

Social Media Text Analytics

- Types of Social Media Text
- Purpose of Text Analytics
- Steps in Text Analytics
- Social Media Text Analysis Tools

Unstructured Data convert to Structured data

Unstructured Data: Customer Feedback Emails

Imagine a company that receives a large volume of customer feedback emails on a daily basis. These emails contain various types of unstructured data, such as:

Email 1:

Subject: Issue with Product Quality

Hello,

I recently purchased your product and I'm quite disappointed with the quality. The material used feels cheap and the product broke within a week of use.

Regards,

John

Unstructured Data convert to Structured data

Email 2:

Subject: Excellent Customer Service

Dear Support Team,

I wanted to commend your customer service representative, Sarah, for her excellent assistance. She went above and beyond to help me resolve my issue and I'm extremely satisfied.

Best regards,

Emily

Unstructured Data convert to Structured data

Structure Data

Email ID	Category	Feedback	Customer Name
-----	-----	-----	-----
001	Product Issue	Poor quality, broke fast	John
002	Service Praise	Excellent help by Sarah	Emily

Social Media Text Analytics

- Text is one of the fundamental elements of social media platforms.
- Textual elements of social media include comments, tweets, blog posts, product reviews, and status updates.
- Text analytics, also known as text mining or text data mining, is the process of extracting meaningful insights and information from unstructured textual data. It involves the application of various techniques from natural language processing (NLP), machine learning, and computational linguistics to analyze, understand, and derive valuable insights from large volumes of text.
- The goal of text analytics is to transform raw textual data, which can be in the form of documents, articles, social media posts, emails, and more, into structured and actionable information.

Social Media Text Analytics

- Review 1: The battery life of the smartphone is terrible. It barely lasts a few hours.
- Review 2: I love the camera quality of this phone! The pictures are so clear and vibrant.
- Review 3: The customer service was a nightmare. They were unhelpful and rude on the phone.
- Review 4: This smartphone is amazing. It's fast, sleek, and the display is gorgeous.

Social Media Text Analytics

Sentiment Analysis:

- Using sentiment analysis, we can determine the sentiment expressed in each review.

Review 1: Negative sentiment (complaint about battery life)

Review 2: Positive sentiment (praise for camera quality)

Review 3: Negative sentiment (complaint about customer service)

Review 4: Positive sentiment (praise for overall performance)

Social Media Text Analytics

Keyword Extraction:

- Extracting keywords from the reviews to identify important topics.

Review 1: battery life, terrible

Review 2: camera quality, clear, vibrant

Review 3: customer service, nightmare, unhelpful, rude

Review 4: smartphone, amazing, fast, sleek, display, gorgeous

Social Media Text Analytics

Topic Modeling:

- Using topic modeling, we can categorize the reviews into topics.

Topic 1: Battery Life (Review 1)

Topic 2: Camera Quality (Review 2)

Topic 3: Customer Service (Review 3)

Topic 4: Overall Performance (Review 4)

Social Media Text Analytics

Text Summarization:

- Generating concise summaries of the reviews.

Review 1 Summary: Negative feedback on battery life.

Review 2 Summary: Positive feedback on camera quality.

Review 3 Summary: Negative experience with customer service.

Review 4 Summary: Positive overall impression of the smartphone.

Social Media Text Analytics

- Social media text analytics, also known as text mining, is a technique to extract, analyze, and interpret hidden business insights from textual elements of social media content.
- Organizations use text analysis techniques to extract hidden valuable meaning, patterns, and structures from the user-generated social media text for business intelligence purposes.
- Text analytics, for example, is useful in gaining a quick and accurate understanding of the emotion and sentiment expressed over social media channels (e.g., tweets or Facebook comments) related to a brand or a new product launch.

Social Media Text Analytics

- The volume and speed at which the comments over social media are generated does not allow for manual reading, and calls for advanced text analysis techniques.
- Text analytics has evolved into a well-established field with roots in variety of domains, including data mining, machine learning, natural language processing, knowledge management, and information retrieval.
- Studies have suggested that approximately 80 percent of data in an organization is textual in nature.

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Social Media Text Analytics

- **Types of Social Media Text**
- Purpose of Text Analytics
- Steps in Text Analytics
- Social Media Text Analysis Tools

TYPES OF SOCIAL MEDIA TEXT

- Based on its nature, social media text can be broadly classified into two categories:
 - 1) **Dynamic text**
 - 2) **Static text**

TYPES OF SOCIAL MEDIA TEXT

1. DYNAMIC TEXT

- Dynamic text is a **real-time social media user-generated text or statement to expresses an opinion** about content or information posted over social media.
- Dynamic text is mostly posed by social media users in response to social, political, economic, personal, cultural, or business issues to express their views and feelings related to it.
- Dynamic text is usually **smaller in length** (e.g., a couple of sentences), **diverse in nature, and is updated or deleted more frequently**.
 - Examples : tweets, Facebook comments, and product reviews.

TYPES OF SOCIAL MEDIA TEXT

Most common dynamic social media texts:

1. Tweets 2. Comments 3. Discussion 4. Conversation 5. Reviews.

1. Tweets

- A tweet is a one hundred forty-character message posted by a Twitter user. A tweet may include text, images, video, or links to other websites. A tweet may also include a hashtag (#).
- Hashtags are used to mark keywords or topics in a tweet, and are organically created by Twitter users as a method to categorize messages. A keyword marked by a hashtag can easily appear in Twitter search, and popular hashtags are often trending topics over Twitter.
- Tweets accumulate over time, carry a timestamp and user information, and mostly appear in descending order; that is, the most recent first. Tweet data provides a valuable source for mining value business insights, including exploring trending topics, measuring brand sentiment, and gathering feedback on new products and services.

TYPES OF SOCIAL MEDIA TEXT

2. Comments

- Social media comments are written (usually short) statements that express opinions about content or information posted over social media.
- While most comments are text only, it can also include images, video, or links to other websites.
- The ability to post comments and participate in social media discourse is the underlying characteristic that distinguishes social media from traditional media (e.g., TV and print).
- Like tweets, social media comments are also a great source for mining valuable business insights from social media. Almost all social media platforms provide commenting features. Comments accumulate over time, carry a timestamp and user information, and mostly appear in descending order; that is, the most recent first.

TYPES OF SOCIAL MEDIA TEXT

3. Discussion

- Discussion takes the form of textual or written conversation or debate about a certain topic, product, or service.
- Mostly, discussions among social media users happen through Internet forums. Internet discussion forums are treelike in structure; that is, a forum can contain a number of sub-forums focused on specific topics or threads.
- In these forums users can post questions and reply to questions posted by other users.
- Discussions accumulate over time, carry a time stamp and user information, and mostly appear in descending order; that is, the most recent first. Vault Network is an example of an Internet forum that focuses on online games.

TYPES OF SOCIAL MEDIA TEXT

4. Conversation

- Social media textual conversation (also known as chatting) is an instant exchange of short written messages between two more people.
- Chatting usually takes a casual form and are carried out through dedicated messaging services/tools.
- A variety of messaging tools have been developed for textual conversation, including desktop-based (e.g., Skype); web-based (e.g., Google and Facebook chat) and mobile-based (e.g., WhatsApp).
- An important point to note here is that most of the social media textual conversation is private in nature and may not be subject to mining.

TYPES OF SOCIAL MEDIA TEXT

5. Reviews

- Reviews are critical evaluations of a product or service performed by customers and experts. They can take both longer and shorter forms.
- Reviews by customers are mostly shorter when compared to formal reviews by experts. Reviews can include textual elements and ratings.
- ProductReview.com.au, for example, is a site devoted to product/service reviews and ratings submitted by customers.
- Product reviews can serve as an excellent source for mining customers' opinions and feelings about a product or service.

TYPES OF SOCIAL MEDIA TEXT

2. STATIC TEXT

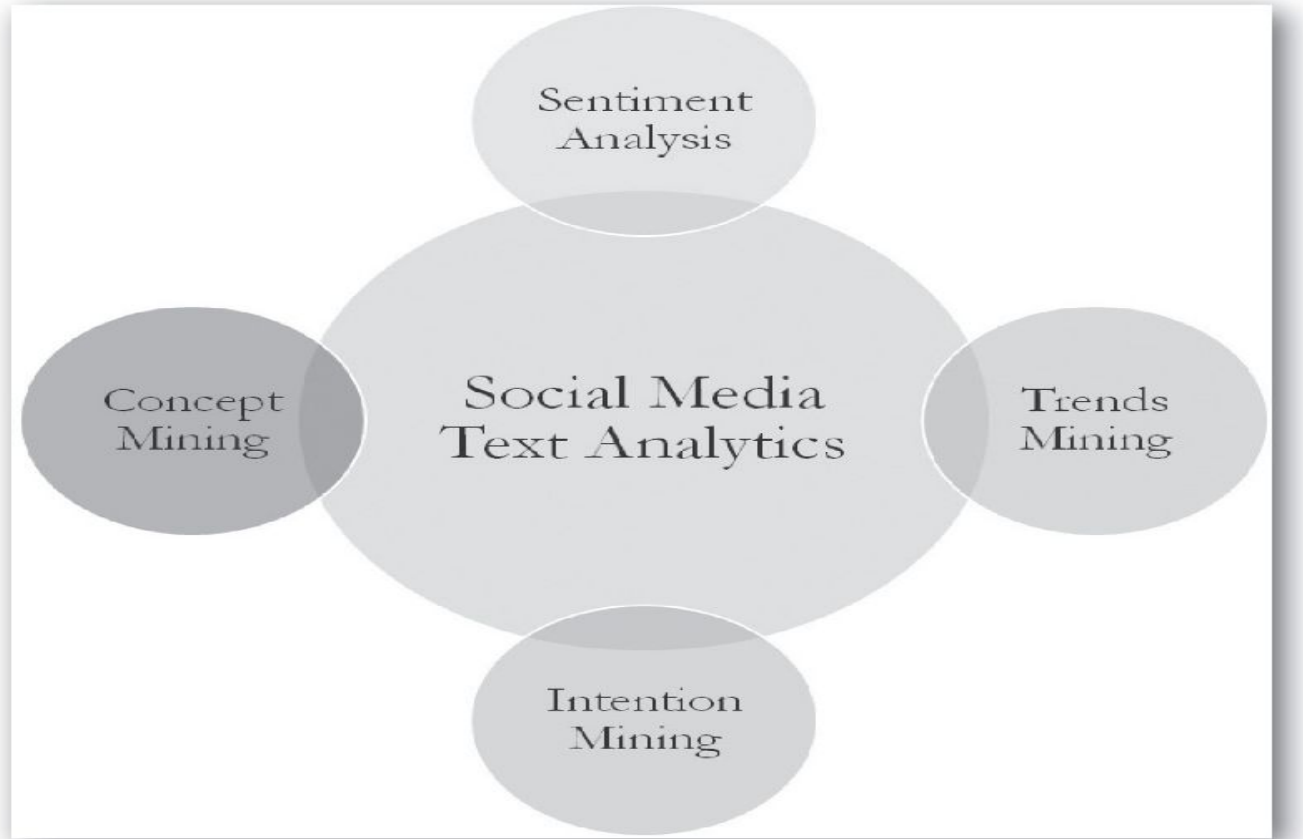
- Static social media text is usually **large in length (e.g., several paragraphs) and is generated, updated, or deleted less frequently.**
- Examples of static text include **wiki content, a blog page, Word documents, corporate reports, electronic mail (e-mail), and news transcripts.**
- At the highest level of abstraction, the **purpose of static social media text is to inform, educate, and elaborate.**

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Social Media Text Analytics

- Types of Social Media Text
- **Purpose of Text Analytics**
- Steps in Text Analytics
- Social Media Text Analysis Tools

PURPOSE OF TEXT ANALYTICS



PURPOSE OF TEXT ANALYTICS

SENTIMENT ANALYSIS

- Sentiment analysis analyzes and categorizes **social media text as being positive, negative, or neutral.**
- Social media sentiment analysis mostly focuses on dynamic text. The primary **purpose of sentiment analysis is to determine how your customers feel about a particular product, service, or issue.**
- For example, as a product manager, you might be interested to know how your customers on Twitter feel about a product/service that was recently launched. Analyzing your tweets or Facebook comments may provide an answer to your question.
- Using sentiment analysis, you may be able to extract the wordings of the comments and determine if they are positive, negative, or neutral.

PURPOSE OF TEXT ANALYTICS

INTENTION MINING

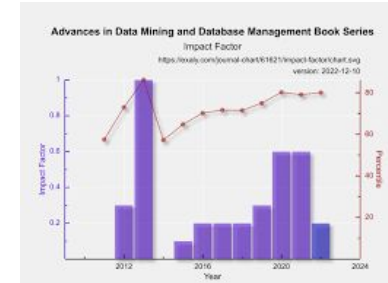
- **Intention or intent mining aims to discover users' intention (such as buy, sell, recommend, quit, desire, or wish) from natural language social media text such as user comments, product reviews, tweets, and blog posts.**
- Social media as the integral part of our contemporary lives and is widely used by millions of customers to express desires, needs, and intention .
- Companies may use intent mining to find new potential customers who intend to buy a product (or services) and service existing customers who have trouble with a product.
- For example, **an analysis of company-related tweets may detect purchase intention based on the presence of the word “buy” or “purchase.”**
- Similarly, **detecting the “quit” intention may identify and service the customers at risk of leaving the company.**



PURPOSE OF TEXT ANALYTICS

TRENDS MINING

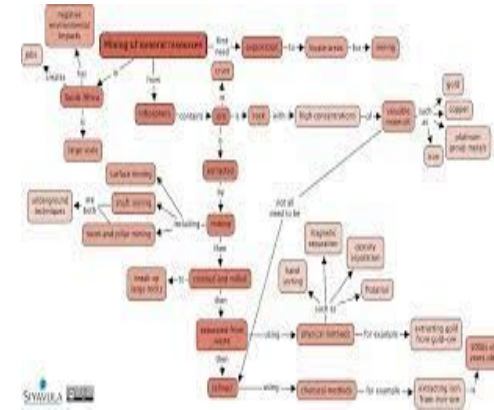
- Trends mining, also known as **predictive analytics**, uses **huge amounts of historical and real-time social media data to predict future events**.
- For example, a vast amount of social media data (e.g., comments and tweets) can be mined to **identify patterns and trends for new product or service development or to improve customer satisfaction by anticipating their needs**.
- Trend mining exploits patterns in large amounts of data by using sophisticated statistical techniques, including machine learning, data mining, and social network analysis.
- Predictive analysis using conventional business data has been used in a variety of domains, including marketing, banking, telecommunication, and healthcare.



PURPOSE OF TEXT ANALYTICS

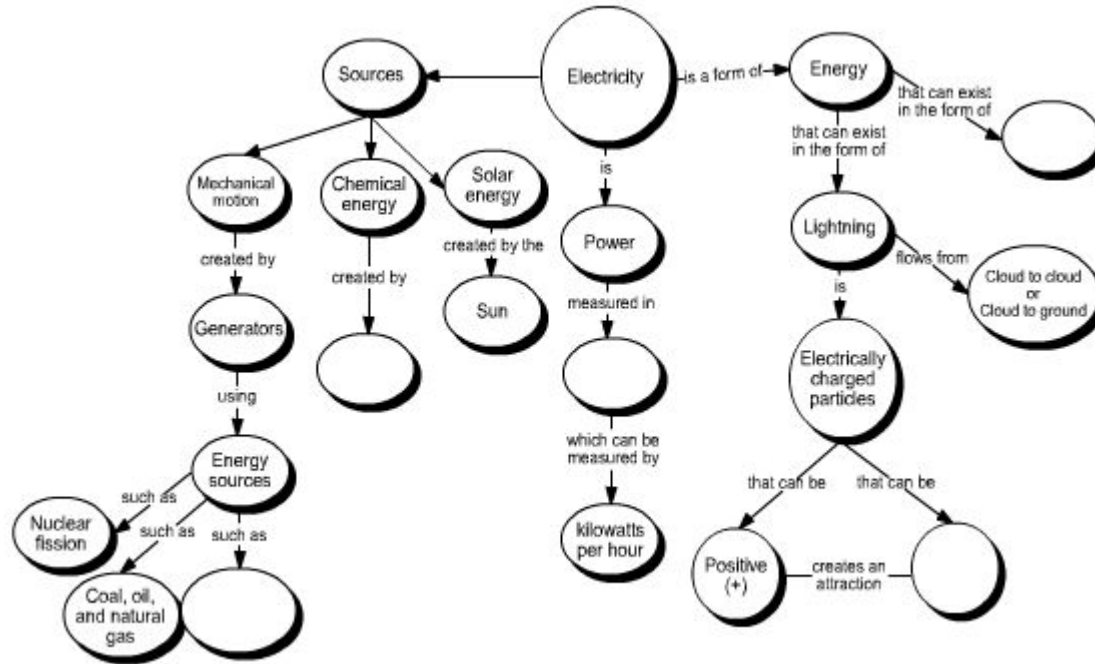
CONCEPT MINING

- Concept mining aims to **extract ideas and concepts from documents.**
- Unlike text mining, which is focused on extracting information, concept mining extracts ideas from large document sets.
 - concept mining is useful in extracting ideas from large amounts of static social media text, such as wiki content, a web page, Word documents, and news transcripts.
- Concept mining can be **employed to classify, cluster, and rank ideas.**



PURPOSE OF TEXT ANALYTICS

CONCEPT MAP



Module 3

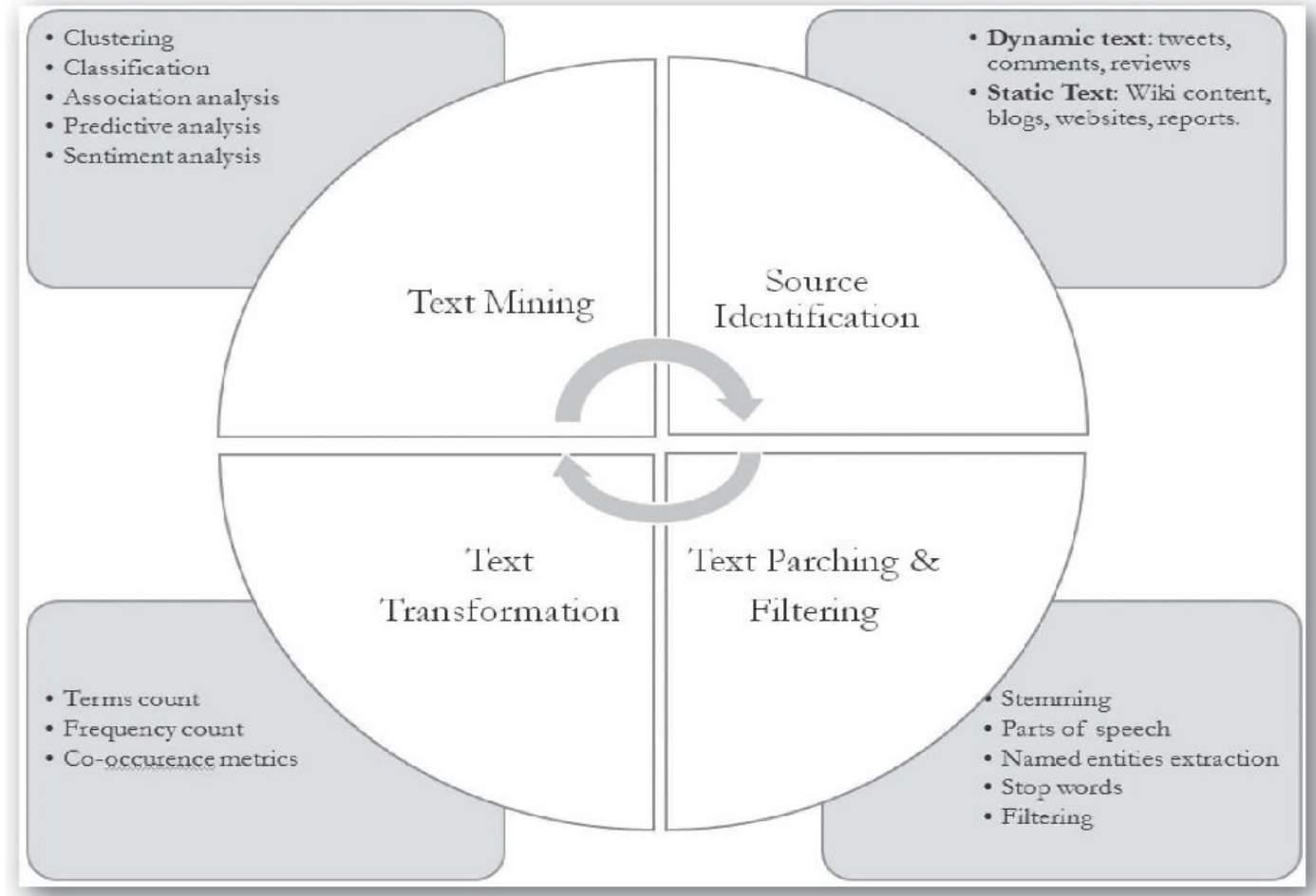
Social Media Text Analytics

- Types of Social Media Text
- Purpose of Text Analytics
- **Steps in Text Analytics**
- Social Media Text Analysis Tools

Steps in Text Analytics

- Text analytics, like any other form of social media analytics, **is the art and science of getting the desired business intelligence from the text posted over social media.**
- While the steps required for text analytics are largely dependent on the type of approach and tool employed, a typical social media text analysis includes the following cyclical steps:
 - Source identification
 - Text parsing and filtering
 - Text transformation
 - Text mining

Steps in Text Analytics



Steps in Text Analytics

IDENTIFICATION AND SEARCHING

- The text analytics process starts with identifying the source of the text that will be analyzed.
- **Text posted on social media is dynamic, huge, diverse, multilingual, and noisy.** Thus, finding the right source for the purpose of text analytics is very crucial for gaining useful business insights.
- The **genre of the source text** also will determine the type of tool used to extract and analyze it. For example, extracting tweets requires different tools and approaches than analyzing a document or website text.
- Analyzing tweets, for example, requires API-based searching and extraction of data from the Twitter timeline based on criteria that you specify. You can choose to extract tweets that include specific keywords, such as your company name.
- The desired business question that needs to be answered with text analytics will serve as a good starting point.

Steps in Text Analytics

TEXT PARSING AND FILTERING

- The next step is to **parse, clean, and filter the text, and create a dictionary of words using NLP**, which is mostly based on machine learning techniques.
- In order for computer and algorithms to extract meanings from the text, **the sentence structures and parts of speech are determined, named entities extracted (people, organizations, product/service names, etc.), stop words removed, and spellings are checked.**
- Most of these steps are automatic; however, in certain stages, human intervention is required. For example, in the filtering stage, manually cleaning (by humans with domain expertise) may be required to remove unwanted or irrelevant terms.

Steps in Text Analytics

TEXT TRANSFORMATION

- For analytical algorithms to be applied to the text, it should be transformed into a computer-readable format (e.g., 0s and 1s) for analysis.
- The cleaned text is thus transformed into numerical representations using linear algebra-based techniques, such as **latent semantic analysis and vector space models**.

Steps in Text Analytics

TEXT MINING

- At this step, the text is actually mined to extract the needed business insights.
- Varieties of text mining algorithms are applied to the text, such as **clustering, association, classification, and predictive analysis, and sentiment analysis.**
- Text analysis employs these sophisticated algorithms to extract sentiment and meanings from the text in a similar manner to the way human do; however, the process is thousands of times faster.

Steps in Text Analytics

TEXT MINING

Association

- Association or association mining is a data-mining technique used to **determine the probability of the co occurrence of items in a collection of documents.**
- The relationships between co occurring items are expressed as association rules.
- In text analytics, for example, social media text can be clustered together based on co occurrence frequency. Or it can be used, for example, **to find that a user who liked a social media content A and B is 90 percent likely to also like content C.**

Steps in Text Analytics

TEXT MINING

Clustering

- Clustering or **cluster analysis groups objects based on similarity in non-overlapping groups.**
- Clustering is an important part of **data mining and text analytics.**
- Social media text (such as tweets or comments), for example, can be clustered into positive, negative, and natural categories. Or nodes in a social media network can be clustered based on importance.

Steps in Text Analytics

TEXT MINING

Classification

- From the text analytics perspective, classification or categorization is used to find similarities in the document and groups them with predefined labels based on the themes contained in the document.
- For example, **an e-mail can be classified as spam based on its contents.**

Steps in Text Analytics

An example of how the four steps of text analytics can be used to extract insights from customer reviews.

- **Source identification:** The first step is to identify the source of the customer reviews. In this case, the source is a website where customers can leave reviews for products and services.
- **Text parsing and filtering:** The next step is to clean and prepare the text data for analysis. This includes removing noise, such as stop words and punctuation, and normalizing the text. In this case, the stop words would be words like "the", "is", and "and". The text would also be normalized by converting all words to lower case.

Steps in Text Analytics

- **Text transformation:** The third step is to convert the text data into a format that can be analyzed by machine learning algorithms. This can involve vectorization, which is the process of representing each word in the text as a vector of features. In this case, the features could be the word frequency, the part of speech, and the sentiment of the word.
- **Text mining:** The final step is to apply machine learning algorithms to the text data to extract insights. In this case, the machine learning algorithm could be used to identify the most common topics in the customer reviews, the sentiment of the reviews, and the main entities mentioned in the reviews.

Steps in Text Analytics

- The insights that can be extracted from customer reviews can be used to improve the product or service, such as by identifying areas where the product or service can be improved or by identifying new features that customers would like.

Steps in Text Analytics

Some other examples of text analytics:

- A company can use text analytics to identify customer complaints and trends in customer sentiment.
- A government agency can use text analytics to monitor social media for signs of unrest or other threats.
- A healthcare provider can use text analytics to identify patients who are at risk for certain diseases.
- A financial institution can use text analytics to detect fraud or money laundering.

Steps in Text Analytics

An example of how a government agency can use text analytics to monitor social media for signs of unrest or other threats:

- **Source identification:** The first step is to identify the sources of social media data that the government agency wants to monitor. This could include popular social media platforms, such as Twitter, Facebook, and Instagram, as well as more niche platforms, such as those used by specific communities or groups.
- **Text parsing and filtering:** The next step is to clean and prepare the social media data for analysis. This includes removing noise, such as stop words and punctuation, and normalizing the text. The stop words would be words like "the", "is", and "and".

The text would also be normalized by converting all words to lower case

Steps in Text Analytics

- **Text transformation:** The third step is to convert the social media data into a format that can be analyzed by machine learning algorithms. This can involve vectorization, which is the process of representing each word in the text as a vector of features. The features could be the word frequency, the part of speech, and the sentiment of the word.
- **Text mining::** The final step is to apply machine learning algorithms to the social media data to extract insights. This could involve tasks such as sentiment analysis, topic modeling, and named entity recognition.

Steps in Text Analytics

For example, a government agency could use sentiment analysis to identify posts that express anger or frustration, which could be a sign of potential unrest. Topic modeling could be used to identify the most common topics being discussed on social media, which could help the government agency to identify potential threats or risks. Named entity recognition could be used to identify people or organizations that are being mentioned in social media posts, which could help the government agency to track potential threats or risks.

Steps in Text Analytics

The insights that can be extracted from social media data can be used by government agencies to prevent or mitigate unrest or other threats. For example, the government agency could use the insights to identify potential hot spots where unrest is likely to occur, or to target specific messages or interventions to address the underlying causes of unrest.

It is important to note that text analytics is not a perfect tool. It can be difficult to accurately identify threats or risks from social media data, and the insights that can be extracted from the data can be subjective and open to interpretation. However, text analytics can be a valuable tool for government agencies that are looking to monitor social media for signs of unrest or other threats.

Module 3

Social Media Text Analytics

- Types of Social Media Text
- Purpose of Text Analytics
- Steps in Text Analytics
- **Social Media Text Analysis Tools**

Social Media Text Analysis Tools

A variety of social media text analysis tools are available on the market. Some are free and others are paid. Below we list some popular text analysis tools.

- **Discovertext:** Discovertext (<http://discovertext.com/>) is a powerful platform for collecting, cleaning, and analyzing text and social media data streams.
- **Lexalytics:** Lexalytics (<http://www.lexalytics.com/>) is a social media text and semantic analysis tool for social media platforms, including Twitter, Facebook, blogs, etc.
- **Tweet Archivist:** Tweet Archivist (<https://www.tweetarchivist.com/>) is focused on searching, archiving, analyzing, and visualizing tweets based on a search term or hashtag (#).

Social Media Text Analysis Tools

- **Twitonomy:** Twitonomy (<https://www.twitonomy.com/>) is a Twitter analytics tool for getting detailed and visual analytics on tweets, retweets, replies, mentions, hashtags, followers, etc.
- **Netlytic:** Netlytic (<https://netlytic.org>) is a cloud-based text and social network analytics platform for social media text that discovers social networks from online conversations on social media sites.
- **LIWC:** Linguistic Inquiry and Word Count (LIWC) is a text analysis tool for analyzing emotional, cognitive, structural, and process components present in individuals' verbal and written speech samples: <http://www.liwc.net/>
- **Voyant:** Voyant (<http://voyant-tools.org/>) is a web-based text reading and analysis. With Voyant, a body of text can be read from a file or directly exported from a website.

Review Questions

- 1. What is text analytics, and why it is useful?**
- 2. Differentiate between static and dynamic social media text.**
- 3. Discuss different social media texts.**
- 4. Explain the four main purposes of social media text analytics.**
- 5. Explain the typical social media text analysis steps.**

Module 3

Social Media Action Analytics

- What Is Actions Analytics?
- Need for Social Media Action Analytics
- Common Social Media Actions
- Actions Analytics Tools

Social Media Action Analytics

- Social media actions analytics deals with extraction, analysis, and interpretation of the insights contained in the **actions performed by social media users**.
- Social media actions are of great value to social media marketers because of their role in **increasing revenue, brand value, and loyalty**.
- Organizations can employ actions analytics to measure popularity and influence of a **product, service, or idea over social media**.
- For example, a brand marketer might be interested to know how popular their new product is among social media users. Analyzing your Facebook **likes and Twitter mentions**, for example, may provide an answer to your questions.

Need for Social Media Action Analytics

- This analysis can be used to **identify trends, measure the impact of social media content, and inform decision-making** about future social media efforts.
- Social media action analytics can **help businesses and organizations understand how their social media presence is perceived and engaged with by users**, and can **provide insights** into the effectiveness of their marketing and communication efforts on social media platforms.
- Social media action analytics is needed to improve customer service.

Common Social Media Actions

- **LIKES**
- **DISLIKES**
- **VIEWS**
- **VISITORS,VISITS,REVISITS**
- **CLICKING**
- **MENTIONS**
- **TAGGING**
- **HOVERING**
- **PINNING**
- **CHECK INS**
- **EMBEDS**
- **UPLOADING AND DOWNLOADING**
- **ENDORSEMENT**

Common Social Media Actions

LIKE

- Like or “Like” buttons or like options are a feature of social media sites (e.g., social networks, blogs, and websites) that **allow users to express their feelings of liking certain products, services, people, ideas, information, places, or content.**
- They are actions performed by **social media users to express symbolic positive reaction to social media content.** Facebook’s “Like” button enables users to easily express their feelings and give your product or service a **virtual thumbs up.**
- Incorporating a “Like” button in social media platforms and websites is becoming a norm. **Social media platforms display accumulated likes received by content over time. Facebook’s “Like” button is the most famous one.**

Common Social Media Actions

DISLIKE

- “Dislike” buttons are included in some **social media platforms (e.g., YouTube)** and **allow users to express their negative feelings of disliking certain content** (e.g., products, services, people, ideas, information, or places) posted over social media.
- Similar to the “Like” feature, it is visible to others and **accumulated over time**.
- The “**Dislike**” button is not as prevalent as the “**Like**” button.
- Some platforms, such as YouTube, allow users to **dislike videos**. This can help creators understand how their content is being received by their audience.

Common Social Media Actions

SHARE

- Share or “Share” button or sharing is a feature that allows social media users to **distribute the content posted over social media to other users.**
- For example, the Facebook “Share” button lets users add a personal message and customize who they share the content with.
- The WordPress (a blogging platform) “Share” button, for example, allows users to share their blog content across a range of social media platforms.
- **Companies incorporate share buttons into website to boost their website traffic by channeling visitors from social media sites.**

Common Social Media Actions

VISITORS, VISITS, REVISITS

- A visitor is a **person who visits your website or blog**. A single visitor may visit a page or content one or more times (revisits). **Visits are also known as sessions.**
- Unique visitor—A person who arrives at your page **first time**.
- Average bounce rate—the **percentage of visitors who visit a website and leave the site quickly without viewing other pages.**
- Session duration—The **average duration of a visit or session**

Common Social Media Actions

VIEWs

- Views are the number of times social media content (a post, video, graphic, etc.) is viewed by users.
- A slightly different but related concept is page views, which is each time a visitor views a page on your company website or blogs
- View data can be used to understand user engagement and popularity of content.

Common Social Media Actions

CLICKS

- Clicks are the actions performed by users by pressing or **clicking on the hyperlinked content of your website or blog.**
- Through clicks, users navigate the web.
- Click data can be harvested for business intelligence purposes, such as, to **reduce bounce rate and improve website traffic.**
- A technique called **clickstream analysis** is used by business managers for a variety of business intelligence purposes, including website activity, website design analysis, path optimization, market research, and finding ways to improve visitor experience on the website.
- **The clickstream is the semistructured data trail/log (such as date and time stamp, IP address, and the URLs of the pages visited) a user leaves while visiting a website.**

Common Social Media Actions

TAGGING

- Tagging is the act of assigning or linking extra pieces of information to social media content (such as **photographs and bookmarks**) for identification, classification, and search purposes.
- Tagging lets user classify social media content the way they see it.
- Tagging may take a variety of forms. For example, bloggers can attach **descriptive keywords (tags) to their posts to facilitate classification and searching of content**, and Facebook users can **add tags to anything they post on their status, including photos and comments**.
- Social bookmarking services (such as Delicious) let users organize their bookmarks flexibly by adding descriptive tags. This practice of collaborative tagging is commonly known as **folksonomy—a term coined by Thomas Vander Wal**.
- These days, almost all prominent companies (e.g., Facebook and Flickr) provide tagging services to their users. Because the contents are tagged with useful keywords, **social tagging expedites the process of searching and finding relevant content**.

Common Social Media Actions

MENTIONS

- Mentions or social mentions are the occurrence of a person, place, or thing over **social media by name.**
- For example, a brand name maybe mentioned in a Facebook comment, blog post, YouTube video, or tweet. **Mentions are important and can indicate popularity of person, place, or thing.**
- For example, a social marketer may be able to gauge the popularity of a product/service/campaign by mining Twitter mentions data. **A Twitter mention is the inclusion of a “@username” in a tweet.**

Common Social Media Actions

HOVERING

- Hovering is the act of moving a cursor over social media content. **Capturing users' cursor movement data can help you understand user behavior on a social media site.**
- Cursor movement/hovering over an ad, for example, can be considered as a **proxy for attention.**
- **Most people who view an ad do not necessary click on it, thus if we are relying on clicks analytics only, we may lose a vital piece of information (i.e., attention).**
- Studies have even suggested a **strong correlation between hover time and purchases.**
- Traditionally, hovering data has been used in website design and for **improvement of user experience.**

Common Social Media Actions

CHECK-IN

- **Check-in is a social media feature** that allows users to announce and share their **arrival at a location**, such as a **hotel, airport, city, or store**.
- Many social media services, including **Facebook provide check-in features**.
- The location of the user is determined using GPS (global positioning system) technology. **Check-in data can, for example, be mined to offer location-based services/products**.

Common Social Media Actions

PINNING

- Pinning is an action performed by **social media users to pin and share interesting content (such as ideas, products, services, and information) using a virtual pin-board platform.**
- Some famous pinning platforms include **Pinterest, Tumblr, StumbleUpon, or Digg.**
- Business can use these **virtual pin boards to share information** and connect with and inspire their customers.
- Hotels and Resorts, for example, use Pinterest to curate travel, food, and luxury lifestyle content to inspire customers.

Common Social Media Actions

EMBEDS

- Embedding is the **act of incorporating social media content (e.g., a link, video, or presentation) into a website or blog**. An embed feature lets users embed interesting content into their personal social media outlets.

ENDORSEMENT

- Endorsement is a features of social media that **lets people endorse and approve other people, products, and services**. For example, **LinkedIn** lets user endorse the skills and qualifications of other people in their network.

Common Social Media Actions

UPLOADING AND DOWNLOADING

- In simple words, uploading is the act of **adding new content (e.g., texts, photos, and videos) to a social media platform.**
- The opposite of uploading is downloading; that is, the act of **receiving data from a social media platform.**
- All most all social media content is created and uploaded by users, which is better known as **user-generated content.**
- For some companies, uploading and downloading is the single most important action to measure.
- For Instagram and Flickr, which are both photo-sharing platforms, the number of photos uploaded daily matters more than anything else.

ACTIONS ANALYTICS TOOLS

- **Hootsuite:** Hootsuite is an easy-to-use online platform that enables you to manage your social media presence across the most popular social networks. Hootsuite offers different plans depending on your business needs and budget: free, pro, or enterprise.
- **SocialMediaMineR:** SocialMediaMineR is a social media analytics tool that takes one or multiple URLs and returns the information about the popularity and reach of the URL(s) on social media, including the number of shares, likes, tweets, pins, and hits on Facebook, Twitter, Pinterest, StumbleUpon, LinkedIn, and Reddit. The tool can be accessed from here: <http://cran.r-project.org/web/packages/SocialMediaMineR/index.html>
- **Lithium:** Lithium (<http://www.lithium.com/>) is a social media management tool that provides a variety of products and services, including social media analytics, marketing, crowd-sourcing, and social media marketing.
- **Google Analytics:** Google Analytics (<http://www.google.com/analytics/>) is an analytical tool offered by Google to track and analyze website traffic. It can also be used for blogs and wiki analytics.

ACTIONS ANALYTICS TOOLS

- **Facebook Insights:** Facebook Insights (<https://www.facebook.com/insights/>) helps Facebook page owners understand and analyze trends within user growth and demographics.
- **Klout:** Klout (<https://klout.com/>) measures your influence across a range of social media channels based on how many people interact with your posts. Your Klout score measures your influence on a scale from one to one hundred.
- **Topsy:** Topsy (<http://topsy.com/>) is similar to Icerocket and Social Mention, with its main focus around social media, especially multimedia sites and blogs.
- **Tweetreach:** This tool helps you measure the number of impressions and reach of hashtags. The tool can be accessed here: <https://tweetreach.com> Kred: Kred helps you measure the influence of a Twitter account: www.kred.com
- **Hashtagify:** This tool measures the influence of hashtags: <http://hashtagify.me>
- **Twtrland:** Twtrland is a social intelligence research tool (<http://twtrland.com/>) for analyzing and visualizes your social footprints.
- **Tweetstats:** using your Twitter username, Tweetstats graphs Twitter stats including tweets per hour, tweets per month, tweet timelines, and reply statistics (<http://www.tweetstats.com>).

Questions

1. Define **social media actions analytics**.
2. Briefly list and define **different actions performed** by social media users.
3. **Why it important to measure actions** performed by social media users?

Module 3

Social Media HyperLink Analytics

- Types of Hyperlinks
- Types of Hyperlink Analytics
- Hyperlink Analytics Tools

Social Media HyperLink Analytics

- Social media hyperlink analytics is the process of tracking and analyzing the performance of links on social media platforms. This can include data such as the number of clicks on a link, the demographics of users who click on a link, and the engagement with the content associated with a link.
- Hyperlinks are clickable elements that direct users to another webpage or resource when clicked. Analyzing the performance of these hyperlinks can provide insights into the effectiveness of social media marketing strategies, content engagement, and audience behavior.

Social Media HyperLink Analytics

- **Hyperlinks** are the pathways of **social media traffic**.
- Hyperlinks are references to web resources (such as a **website, document, and files**) that users can access by clicking on it. Hyperlinks can link resources **within a document (inter-linking)** and **among documents (intralinking)**.
- For example, clicking on a hyperlink in a tweet can link you to other resources (e.g., websites) available over the Internet. Hyperlinks are not merely technical links between two websites, but serve a more symbolic means.
- As a website is an official and unique entity representing an organization itself; therefore, embedding hyperlinks in an organization's website can be considered an **official act of communication between two organizations**.

Social Media HyperLink Analytics

- Here are some key aspects of social media hyperlink analytics:
- **Click-through Rate (CTR):** This metric measures the percentage of users who clicked on a hyperlink out of the total number of users who saw the link. It helps gauge the effectiveness of a link in attracting user attention and encouraging clicks.
- **Conversion Tracking:** By integrating tracking parameters (UTM parameters) into hyperlinks, marketers can monitor the conversion of users who clicked on a link and completed a desired action, such as making a purchase or signing up for a newsletter.
- **Referral Traffic:** Social media hyperlink analytics can help identify which social platforms are driving the most traffic to your website. This insight can guide your social media strategy and focus resources on platforms that provide the most value.

Social Media HyperLink Analytics

- **Time and Location Data:** Some analytics tools provide information about when and where users clicked on a hyperlink. This data can help refine posting schedules and tailor content to different time zones or regions.
- **Audience Insights:** Analyzing the demographics and interests of users who clicked on hyperlinks can help refine your target audience and tailor content accordingly.
- **Platform-Specific Metrics:** Different social media platforms may provide varying levels of analytics. For instance, Twitter Analytics, Facebook Insights, and Instagram Insights offer platform-specific data to help you understand how your hyperlinks are performing on each platform.

Social Media HyperLink Analytics

- **Shortened Links:** Many social media platforms automatically shorten links to conserve character space. It's important to use link shortening services that also provide tracking capabilities, allowing you to monitor the performance of these links.

HYPERLINK ANALYTICS

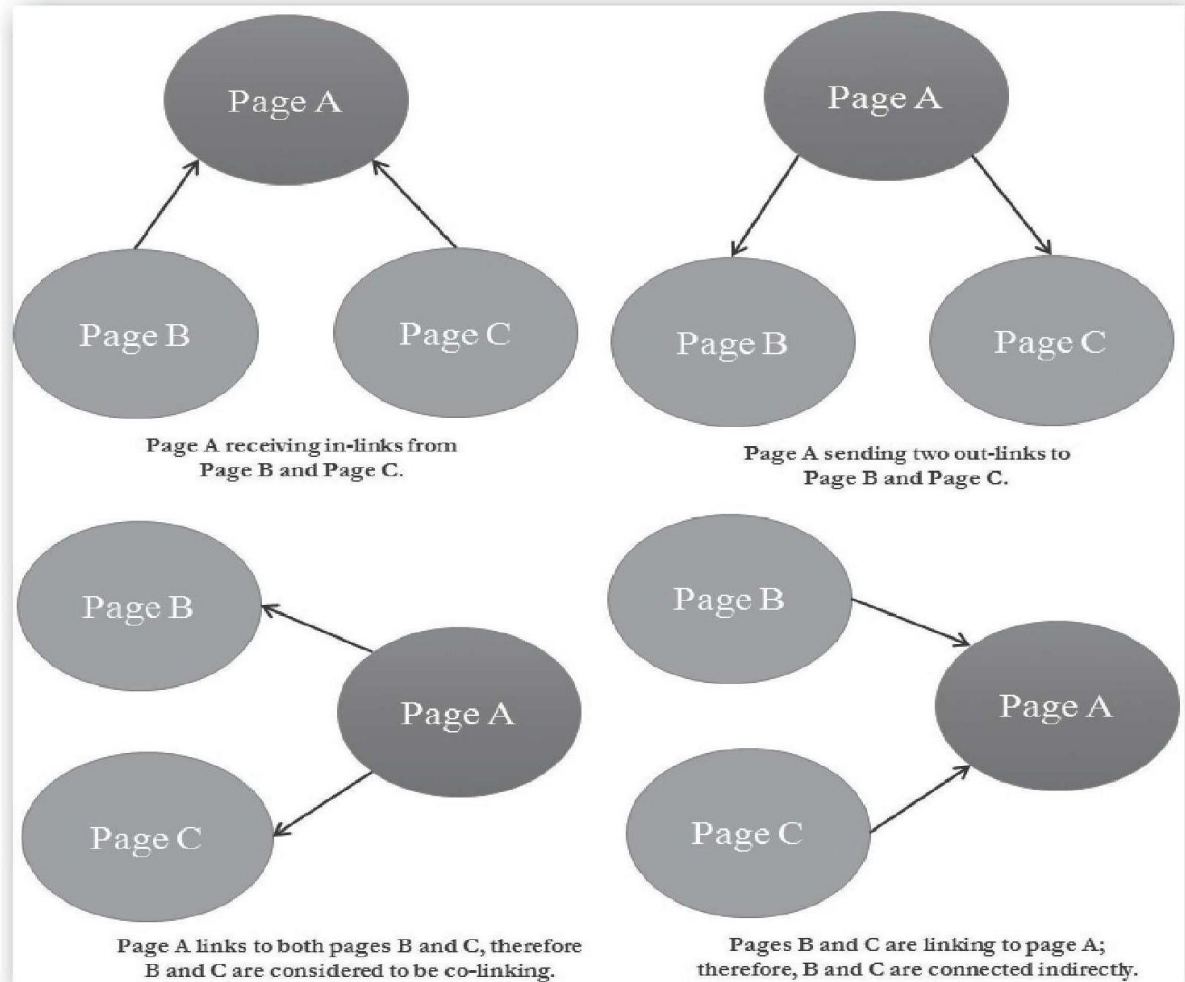
- Hyperlink analytics deals with **extracting, analyzing, and interpreting hyperlinks (e.g., in-links, out-links, and co-links)**. The basic assumption pertaining to hyperlink analytics is that the **number and quality of hyperlinks to a website equates to its importance or value**.
- Hyperlink analytics can also reveal, for example, **Internet traffic patterns and sources of the incoming or outgoing traffic to and from a website**.

Social Media HyperLink Analytics

- Hyperlinks among websites represent not only a reasonable approximation of a **social relationship**, but also serve as a **symbolic meaning of validating or endorsing the linked organization**.
- These hyperlinks that exist between two organizational websites reflect a sense of **validation, trust, bonding, authority, and legitimacy**. Websites mostly connect or link to other websites of similar nature, so **hyperlinks can also serves as indicators of content similarity**.
- From hyperlink analytics point of view, mainly there are three types of hyperlinks,
1) in-links, 2) out-links, and 3) co-links.

Types of Hyperlinks

- **IN-LINKS**
- **OUT-LINKS**
- **CO-LINKS**



Types of Hyperlinks

- **IN-LINKS**
 - In-links are the incoming hyperlinks or links directed toward a website or originated in other websites
- **OUT-LINKS**
 - Out-links are hyperlinks generated **out of a website**
- **CO-LINKS**
 - Co-links have two dimensions.

Social Media HyperLink Analytics

IN-LINKS

- **In-links are the incoming hyperlinks or links directed toward a website or originated in other websites.**
- **In-links are of great interest to social markers, because they bring traffic to a particular website.**
- In-links also play an important role in **website analytics**, as both the quality and number of in-links can impact the **search engine ranking of the website**.
- In-links can also **impact the popularity of social media contents**.
- A study on YouTube viral videos, for instance, found that among other things, in-links play crucial roles in the viral phenomenon, particularly in increasing views of videos posted on YouTube. Studies have also shown that **in-link counts strongly correlate with measures describing business performance**

Social Media HyperLink Analytics

OUT-LINKS

- Out-links are hyperlinks generated out of a website.

CO-LINKS

- Co-links have two dimensions.
- First, if two websites receive a link from a third website, they are considered to be connected indirectly.
 - For example, page A links to both pages B and C, therefore **B and C are considered to be co-linking, or connected indirectly**
 - Second, if two pages link to a third page, they are also considered to be co- linking. Pages B and C are linking to page A; therefore, B and C are connected indirectly.
 - Co-links have been used to **compare and map competitive similarity among companies.**

HYPERLINK ANALYTICS

Hyperlink analysis **uses** and **limitations**:

- **Uses:** Hyperlink analysis has been used to study a variety of topics
 - including **ranking of universities, understanding the blogosphere, scholarly websites , and political networks, and to measure business competitiveness.**
- **Limitations:**

Hyperlink networks **does not**

- **Provide any insight into the type or amount of traffic flowing among websites.**
- **Include hyperlinks within a website between pages.**
- Hyperlinks between pages within a website are created mostly for navigational purposes. Also, search engine ranking algorithms either ignore or **give low importance to hyperlinks within a website.**

TYPES OF HYPERLINK ANALYTICS

TYPES OF HYPERLINK ANALYTICS

Hyperlink analytics can take several forms, including:

- 1) Hyperlink Environment Analysis**
- 2) Link Impact Analysis**
- 3) Social Media Hyperlink Analysis**

TYPES OF HYPERLINK ANALYTICS

Hyperlink environment analysis

- Analysis Hyperlink environment analyses deal with a particular website or set of websites.
- Hyperlinks (i.e., out-links, in-links, and co-links) of a website are extracted and analyzed to identify the sources of Internet traffic.
- Hyperlinks environment networks can take two forms:
 - 1) co-links networks or
 - 2) in-links and out-links networks.

TYPES OF HYPERLINK ANALYTICS

Co-Link Networks

- In co-links environment networks, **nodes are websites** and **links that represent similarity between websites**, as measured by **co-link counts**.

In-Links and Out-Links Networks

- In-links and out-links hyperlink environment networks are constructed based on in-links and out-links from a website or set of websites.
- In such a network, **nodes will be websites** and **links will present in-links and out-links**.

TYPES OF HYPERLINK ANALYTICS

Link impact analysis

- Link impact analysis **investigates the web impact of a website address (or URL) in terms of citations or mentions** it receives over the web.
- In a link impact analysis, **statistics about web pages that mention the URL of a given website are collected and analyzed.**
- The assumption is that a **URL (or website address) cited frequently over the web is more important.** Thus, measuring the web impact of URLs may provide an idea about the **importance of a website.**

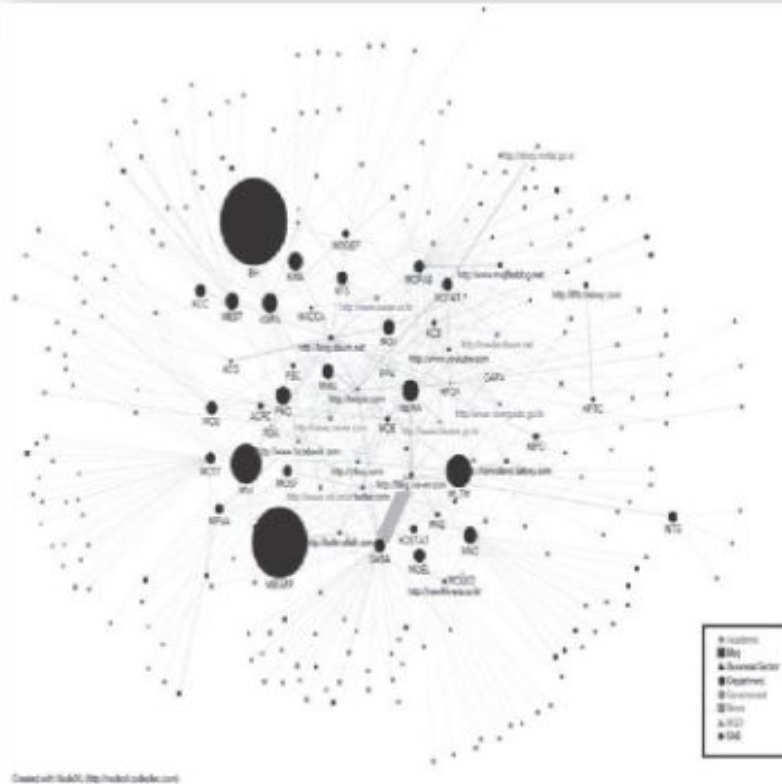
TYPES OF HYPERLINK ANALYTICS

- **Social media hyperlink analysis** deals with extraction and analysis of hyperlinks embedded within social media texts (.e.g., tweets and comments).
- These hyperlinks can be extracted and studied to identify the sources and destination of social media traffic.
- A good example of the usefulness of the hyperlink embedded in the social media text , in which they extracted **out-links from Korean and US government agencies' tweets**.
- By extracting out-links and tracing them back to their sender, the authors were able to constructed a map of the out-link structure.

Social Media Hyperlink Analysis

- According to a comparison of **out-links between tweets of the Korean and US governments**, there were some **differences in citation (i.e., out-link) patterns**.
- The **Korean government tended to cite domestic portals' news services** and their own blogs (i.e., self-citation). Although there were SNSs and newspaper sites, most of the related out-links were for portals.
- On the other hand, the **US government showed a more diverse pattern in terms of out-link destinations**. US out-links were not concentrated in specific sites and tended to go directly to news agencies, not to secondary sources such as portals.
- These comparisons between the **US and Korean governments suggest that social media out-links can carry valuable information and can help explain real-world phenomena and shed light on the disparities in social media use among different cultures**

Social Media Hyperlink Analysis



Twitter out-link diagram for Korean government agencies



Twitter out-link diagram for US government agencies

Case Study - Viral Videos hyperlink network

Top 100 viral youtube videos

- The **squares** indicate **unique videos IDs** and **circle nodes** indicate the **domain names**.
- The size of a node indicates the number of URLs (in-links) received by the videos
- **viral marketers**- in-links should be taken into account while formulating virality strategies.

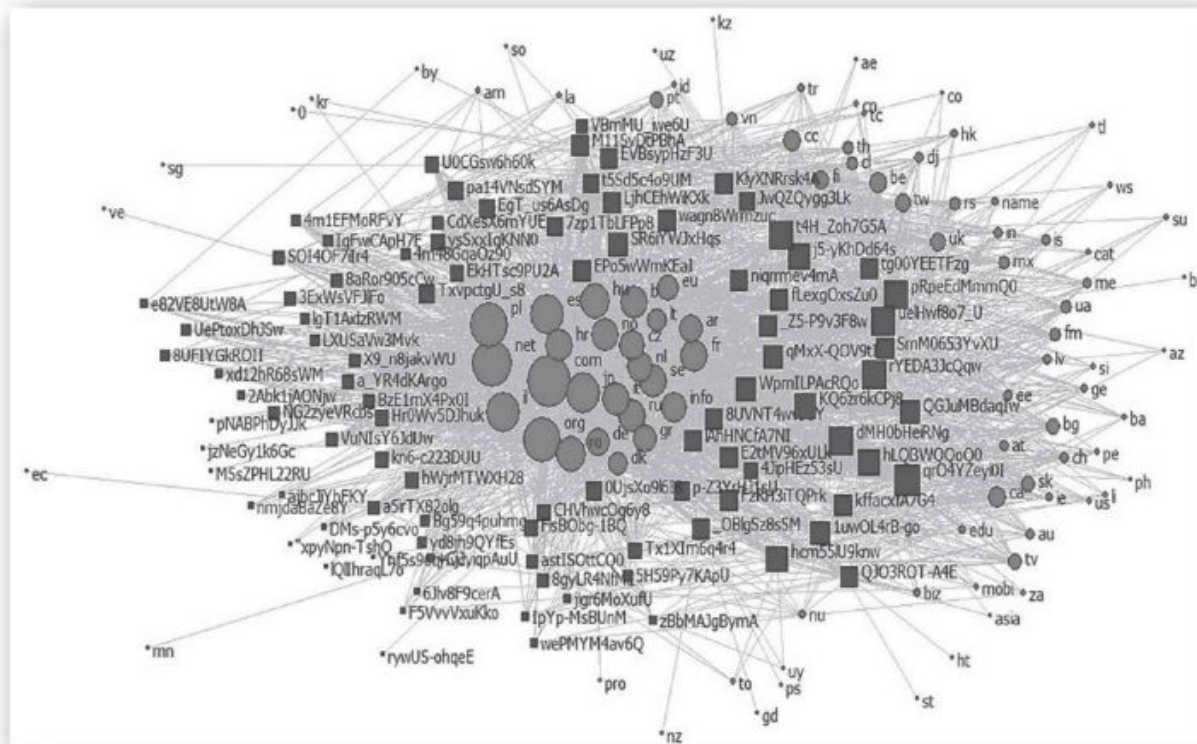


Figure 11. Viral Videos hyperlink network

HYPERLINK ANALYTICS TOOLS

- **Webometric Analyst:** Webometric Analyst is a web impact analysis tool and can conduct variety of analysis on social media platforms including hyperlink network analysis and web mentions: <http://lexiurl.wlv.ac.uk/>
- **VOSON:** VOSON (<http://www.uberlink.com/>) is a hyperlink analytics tools for constructing and analyzing hyperlink networks.
- **Open Site Explorer:** Open Site Explorer is a link analysis tool to research and compare competitor backlinks, identify top pages, view social activity data, and analyze anchor text: <https://moz.com/researchtools/ose/>
- **Link Diagnosis:** Link Diagnosis (<http://www.linkdiagnosis.com/>) is a free online tool for analyzing and diagnosing links.
- <https://smallseotools.com>

HYPERLINK ANALYTICS TOOLS

- **Advanced Link Manager:** Advanced Link Manager provides a variety of link analysis capabilities, including the ability to track link-building progress over time, domain quality analysis, backlinks evolution, and website-crawling abilities: <http://www.advancedlinkmanager.com/>
- **Majestic:** Majestic (<https://majestic.com>) provides a variety of link analysis tools, including link explorer, backlinks history, and link mapping tools.
- **Backlink Watch:** Backlink Watch (<http://backlinkwatch.com/>) is a free tool for checking the quality and quantity of in-links pointing to a website.

Questions

- What are hyperlinks, and why they are important?
- Briefly discuss in-links, out-links, and co-links.
- What is hyperlink analytics and its underlying assumptions?
- What is hyperlink environment analysis?
- What is link impact analysis?
- What is social media hyperlink analysis?

Thank you